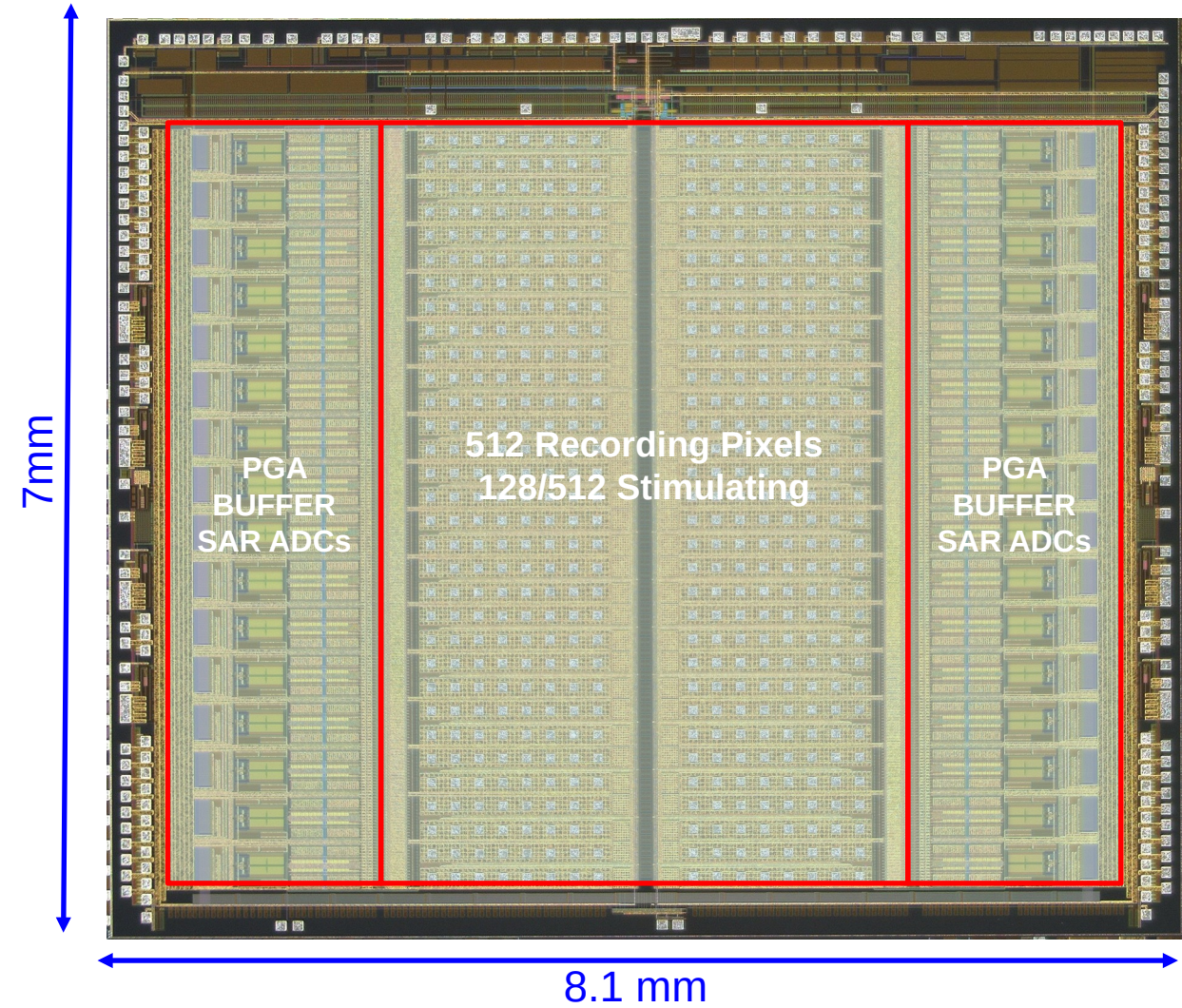
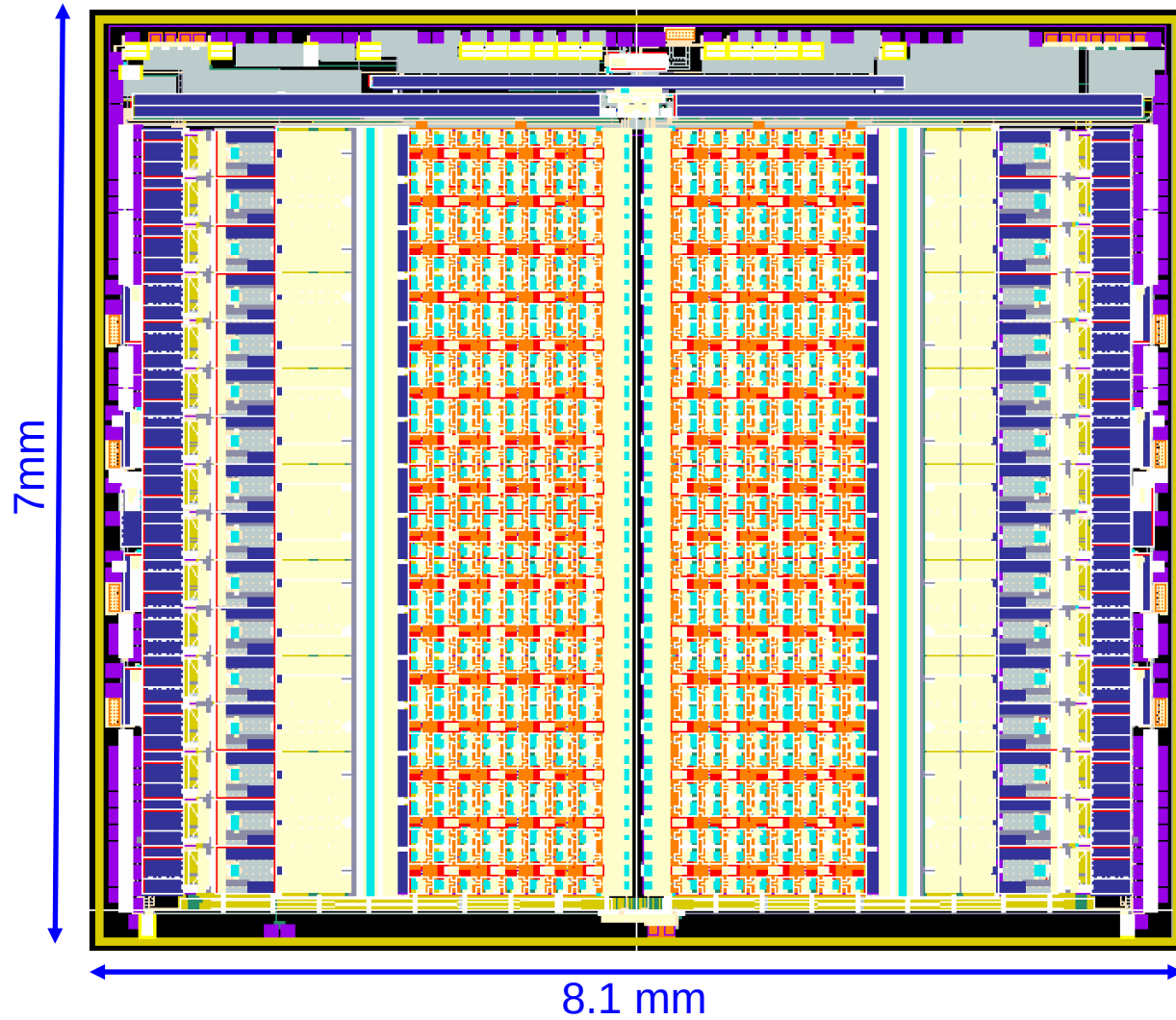
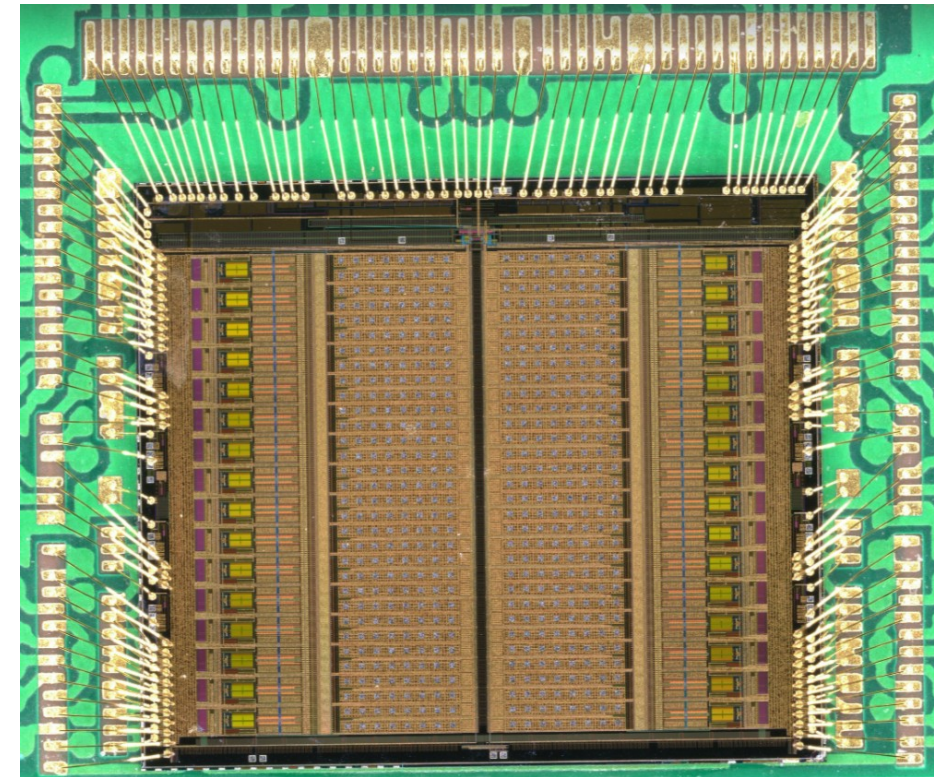
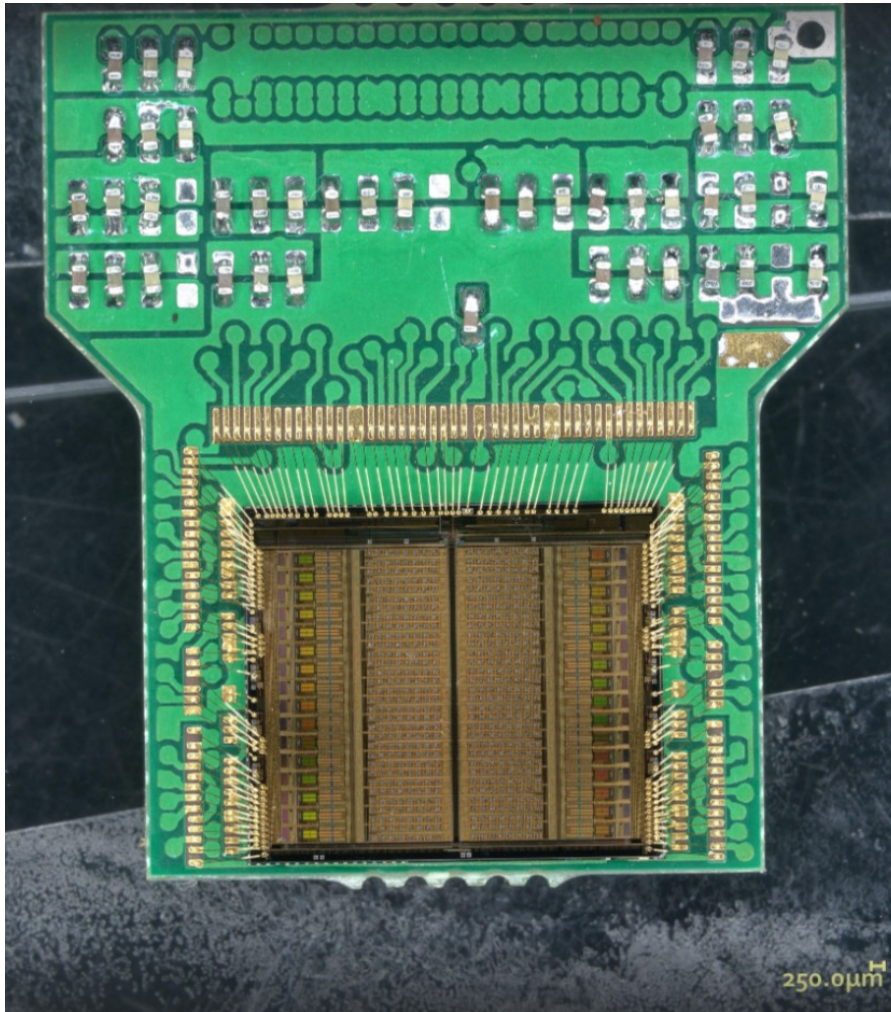


ASIC Layout



Bonding Map

- Bonding recipe for the ASIC-board connection



Chip Specifications

- **Gain**
 - **Programmable 3bits:** (110 – 700)
 - **Raw data bandwidth:** (0.3Hz – 10kHz)
 - **Input Referred Noise:** (4 – 5 μ Vrms)
 - **Sampling rate:** (31.25kS/s)

Neural Recording Chip Specifications	
Parameter	Value
Recording	
Channels	512
Gain (V/V)	110–700
Bandwidth (Hz)	0.3–10k
Input-Referred Noise	4–5 μ Vrms
ADC Resolution	12-bit
Reset Time	< 1.5 ms
Power / Channel	<20 μ W
Input Impedance	13 M Ω @ 1 kHz
Sampling Rate	31.25 kS/s per channel

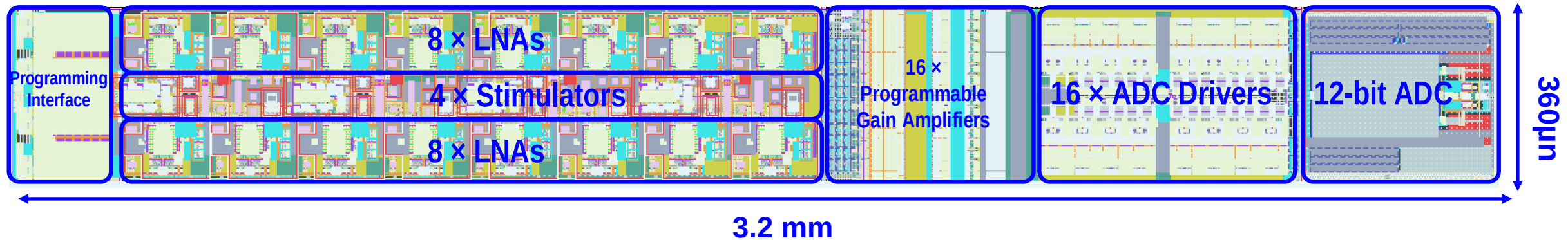
Chip Specifications

- **Gain**
 - **Programmable 3bits:** (110 – 700)
 - **Raw data bandwidth:** (0.3Hz – 10kHz)
 - **Input Referred Noise:** (4 – 5 μ Vrms)
 - **Sampling rate:** (31.25kS/s)
- **Stimulator**
 - **128 channels:** Compliance of +/-10V
 - **Resolution:** 1.5 μ A
 - **Maximum Current:** 22.5 μ A

Neural Recording Chip Specifications	
Parameter	Value
Recording	
Channels	512
Gain (V/V)	110–700
Bandwidth (Hz)	0.3–10k
Input-Referred Noise	4–5 μ Vrms
ADC Resolution	12-bit
Reset Time	< 1.5 ms
Power / Channel	<20 μ W
Input Impedance	13 M Ω @ 1 kHz
Sampling Rate	31.25 kS/s per channel
Stimulator	
Channels	128
Compliance Voltage	$\pm$ 10 V
Max Stimulation Current	22.5 μ A
Stimulation Resolution	1.5 μ A
Stimulation Frequency	< 200 Hz
Temporal Resolution	31.25 μ s
Stimulation Mode	Arbitrary Monopolar / Bipolar
Charge Balancing	< 10 mV residual

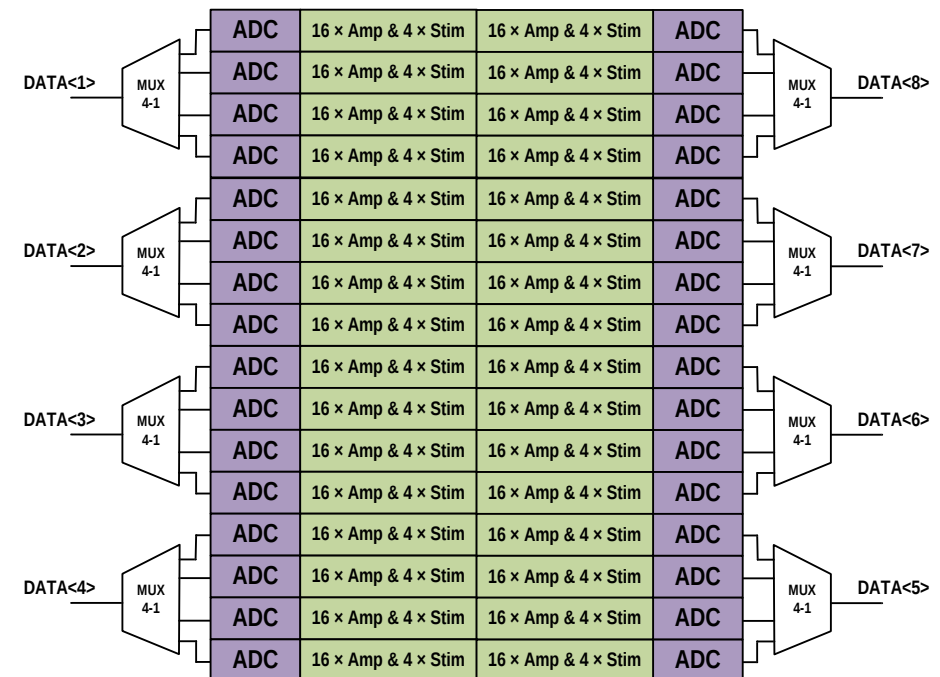


Recording and Stimulation



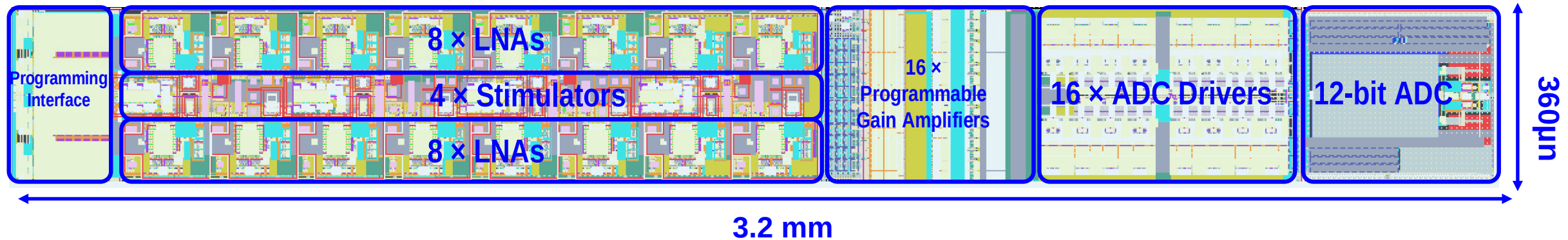
- **Modular concept**

- One module: 16 amplifiers + 4 stimulators
- Arrange 32 copies to make it 512 recording + 128 stimulators
- Programming interface programs the stimulators and the amplifiers
- The ADCs are multiplexed to 8 data lines





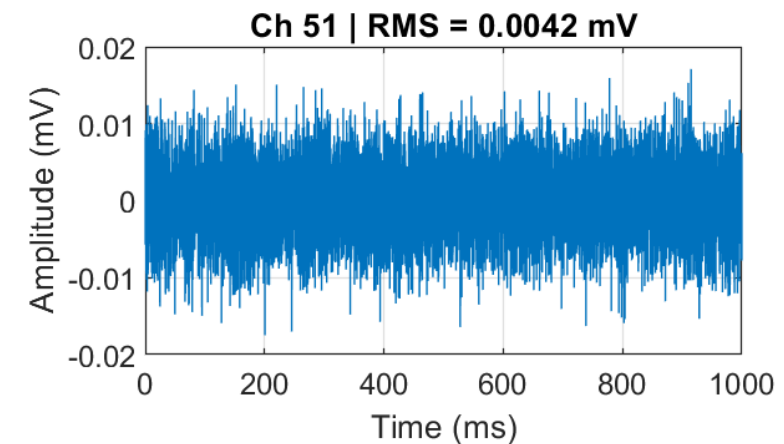
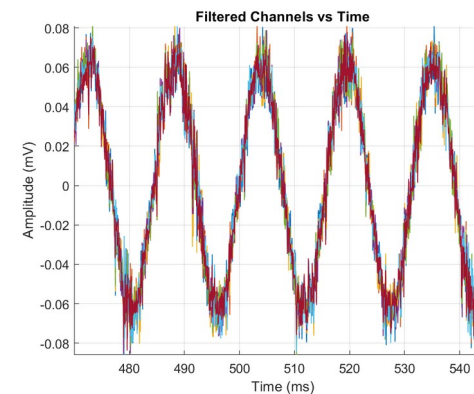
Recording Mode



- **From Pixel Input to ADC Output**

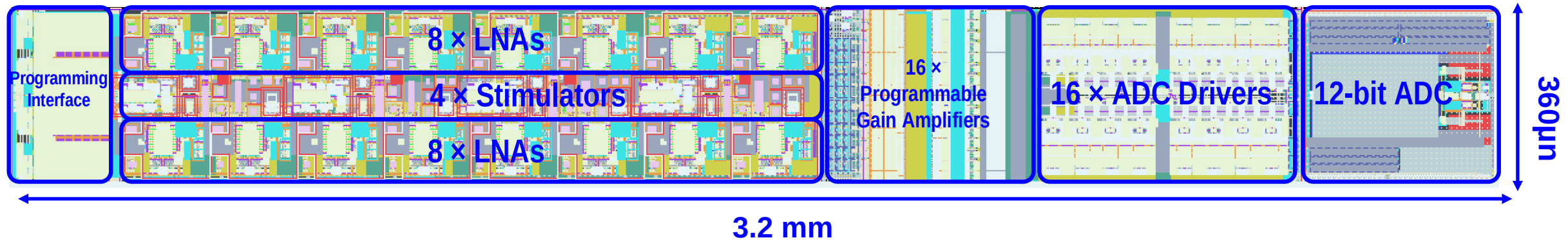
- Gain: 100 – 700 (V/V)
- Bandwidth: ~0.3 Hz to 10 kHz
- IRN = 4 – 5 μVrms
- Sampling Rate: 31.25 kS/s per channel

IRN (Input-Referred Noise): $\sim 4.2 \mu\text{Vrms}$



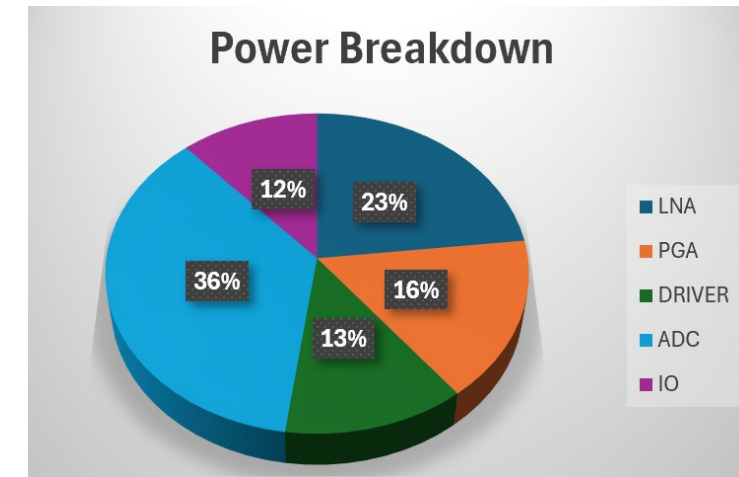


Recording Mode

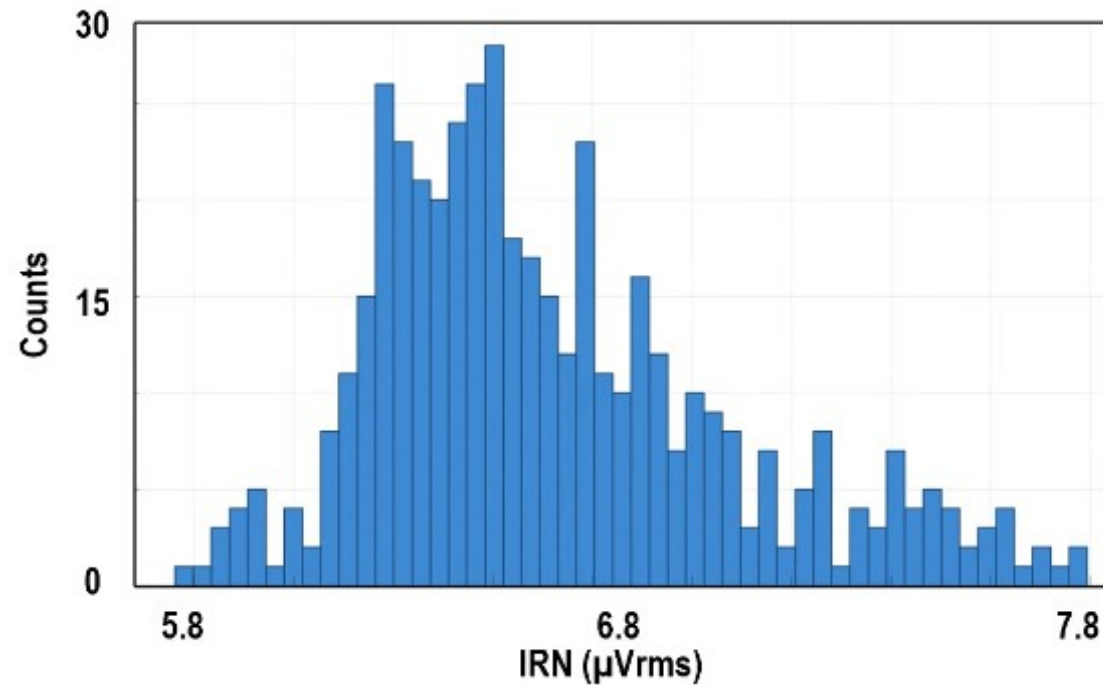
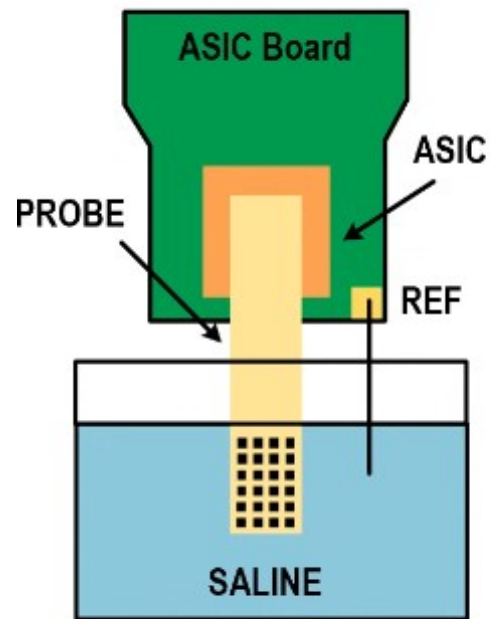


- **From Pixel Input to ADC Output**

- Gain: 100 – 700 (V/V)
- Bandwidth: ~0.3 Hz to 10 kHz
- IRN = 4 – 5 μVrms
- Sampling Rate: 31.25 kS/s per channel
- DC Power: ~17.24 μW / channel
 - Not including the external LDOs



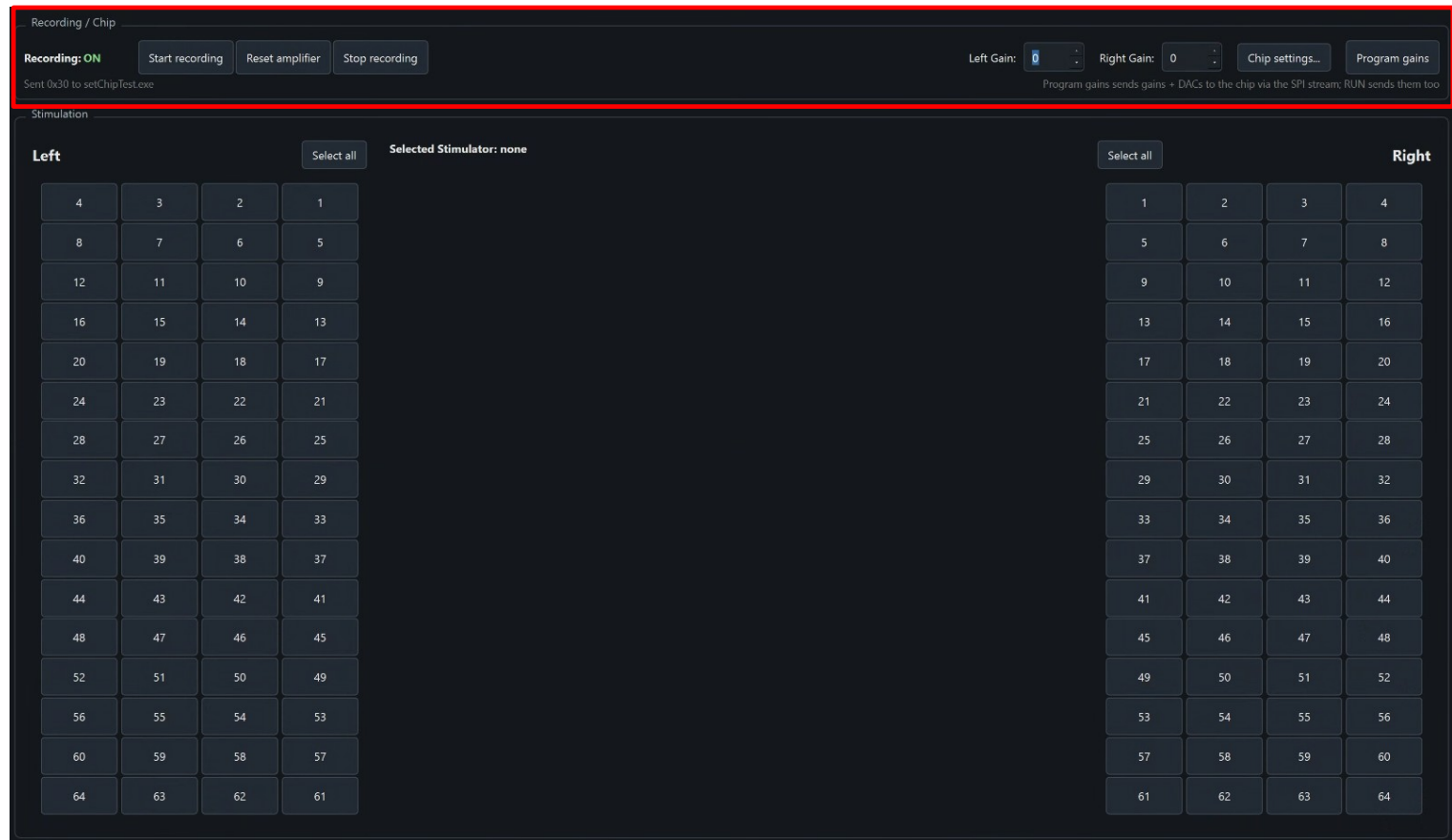
Recording Mode



- Integrated ASIC + Probe noise
- $\mu = 6.5\mu\text{VRMS}$ and $\sigma = 0.3\mu\text{VRMS}$

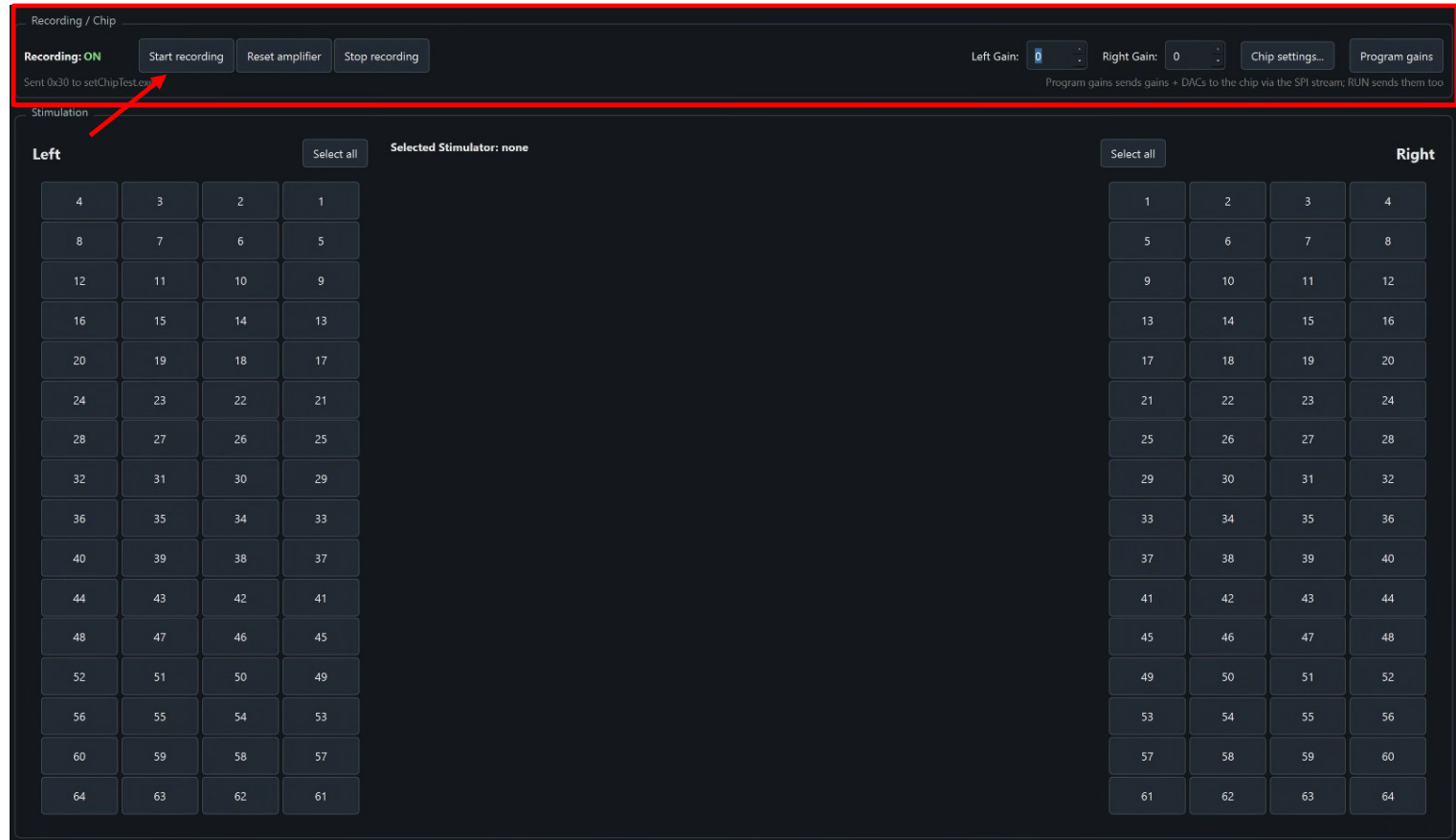
Reading Mode – How?

- Custom GUI
 - Recording on the top
 - Stimulation on the bottom



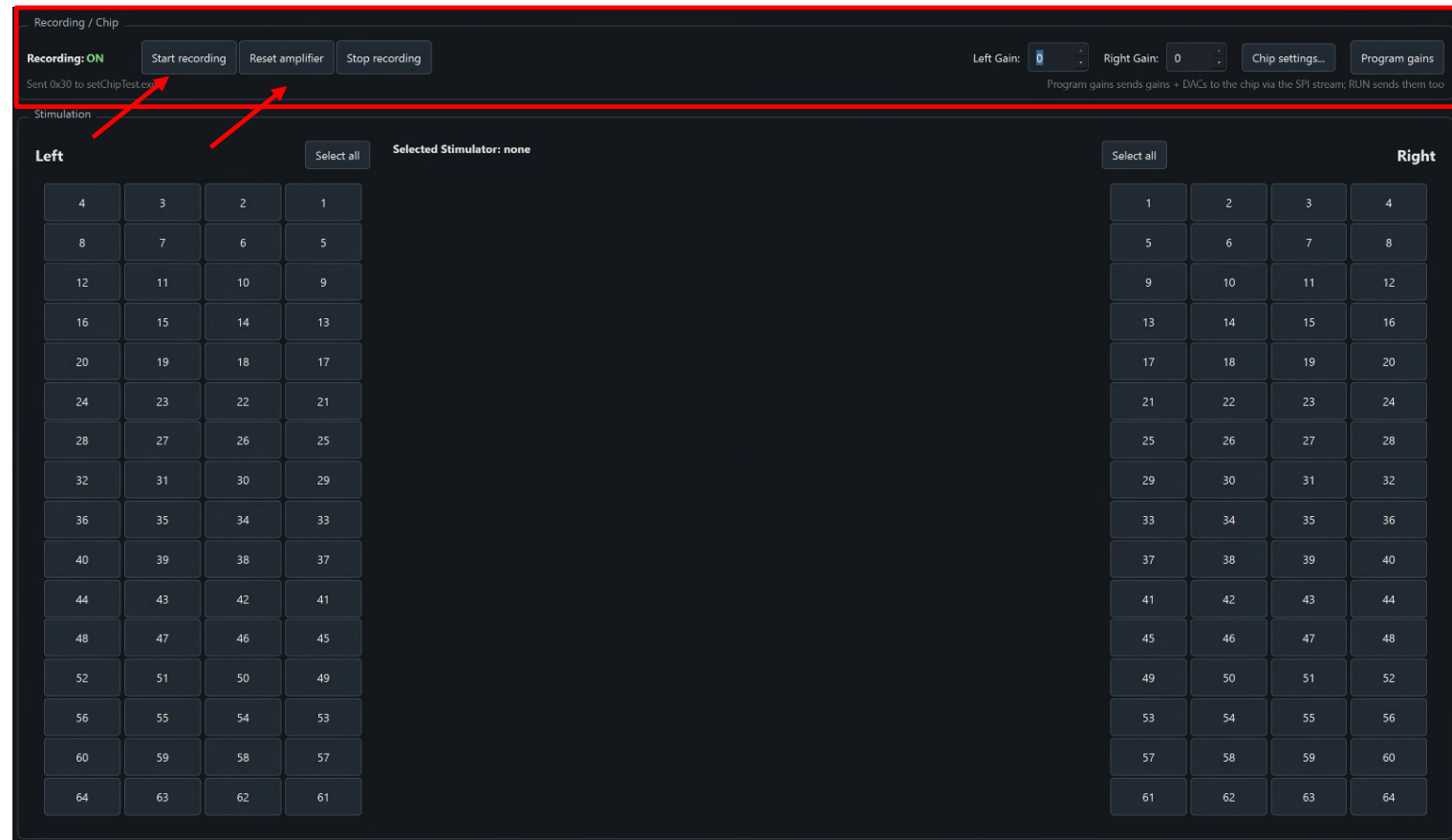
Reading Mode – How?

- **Custom GUI**
 - Recording on the top
 - Stimulation on the bottom
- **Start recording**
 - Sends a sequence to FPGA to trigger the recording



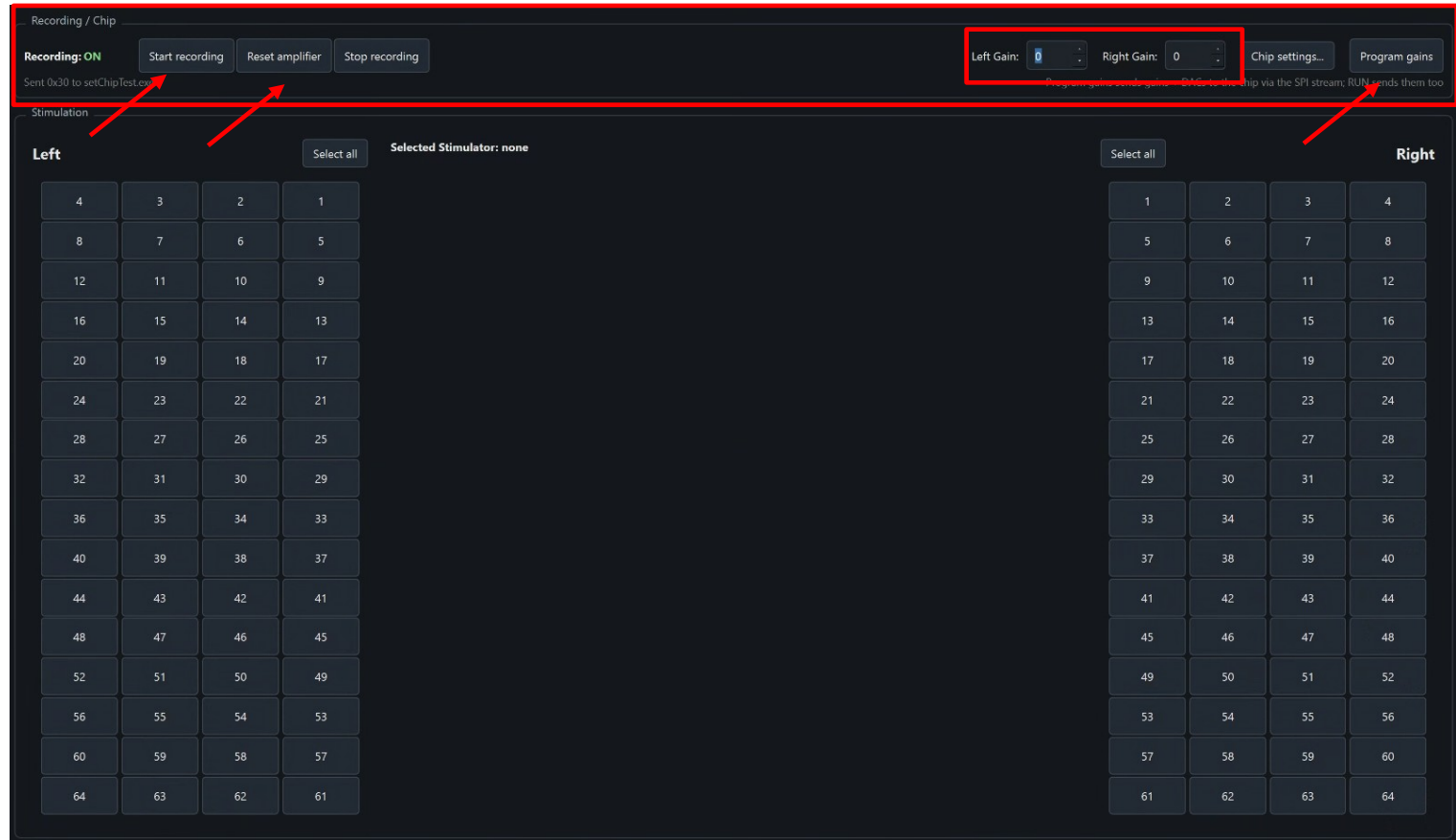
Reading Mode – How?

- **Custom GUI**
 - Recording on the top
 - Stimulation on the bottom
- **Start recording**
 - Sends a sequence to FPGA to trigger the recording
- **Reset amplifier**
 - It sends a sequence to FPGA to trigger the reset the amplifiers



Reading Mode – How?

- **Custom GUI**
 - Recording on the top
 - Stimulation on the bottom
- **Start recording**
 - Sends a sequence to FPGA to trigger the recording
- **Reset amplifier**
 - It sends a sequence to FPGA to trigger the reset the amplifiers
- **Left and Right Gain**
 - They control the gains of the two sides of the chip (they can be different if needed)
 - Gain will be programmed only if Program Gains is pressed

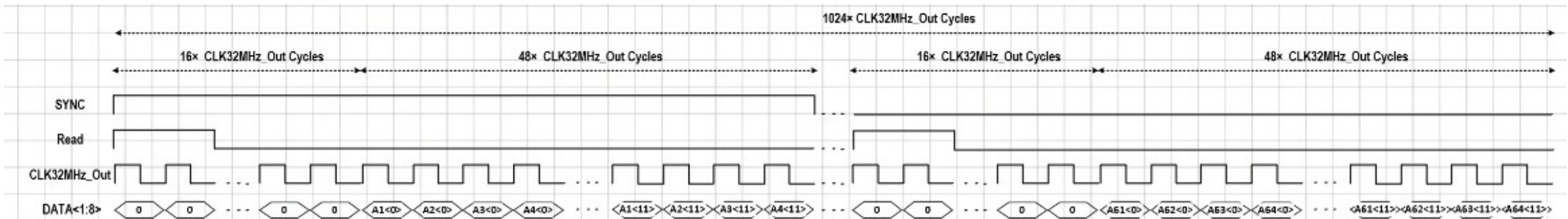


What happens in the background

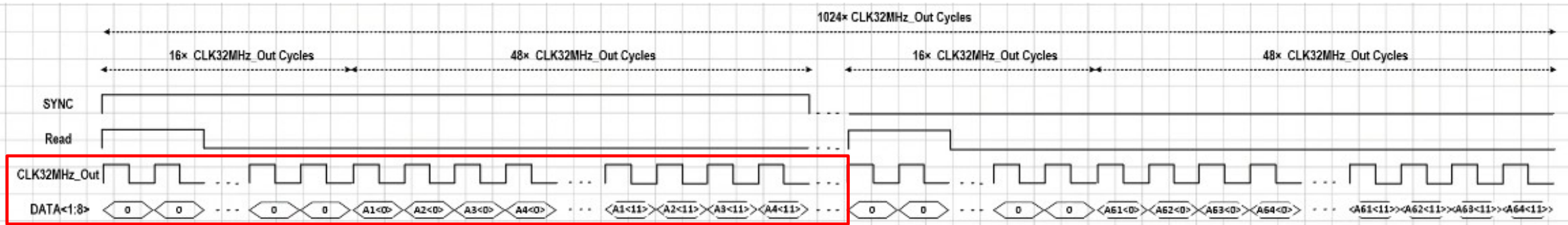
- **Recording Related Pins**
 - **Chip_Reset:** We use this pin to reset the chip to a known state
 - **CLK32MHz_In:** If we press start recording button, the FPGA sends this clock to the ASIC to start the recording
 - **FE_RESET:** If we press reset amplifier button, the FPGA resets the amplifiers we should see very silent channels
 - **Amplifier Gain:** Amplifier gain is set by a Serial Peripheral Interface (SPI will talk more about this later)

What happens in the background

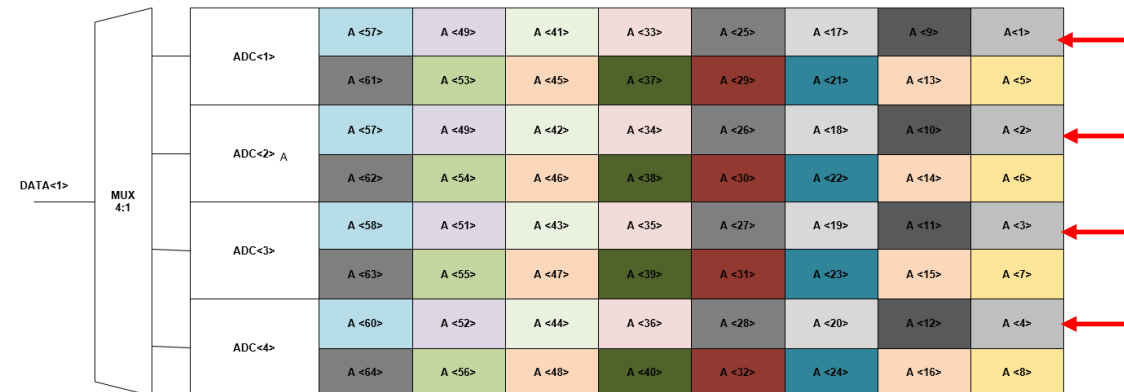
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 - **Chip_Reset:** We use this pin to reset the chip to a known state
 - **CLK32MHz_In:** If we press start recording button, the FPGA sends this clock to the ASIC to start the recording
 - **FE_RESET:** If we press reset amplifier button, the FPGA resets the amplifiers we should see very silent channels
 - **Amplifier Gain:** Amplifier gain is set by a Serial Peripheral Interface (SPI will talk more about this later)
- **When recording starts: Chip will transmit these signals**
 - **CLK32MHz_Out:** Used by FPGA to readout the correct data
 - **Data<1:8>:** Contains the recording data
 - Each data line contains the information of 4 ADC (time multiplexed)
 - One ADC → 16amps, then 1 Data → 64 amps.
 - **Read:** Used to let the FPGA know that a new amplifier's data is being streamed
 - **Sync:** Used to let the FPGA know which amplifier's data is being streamed



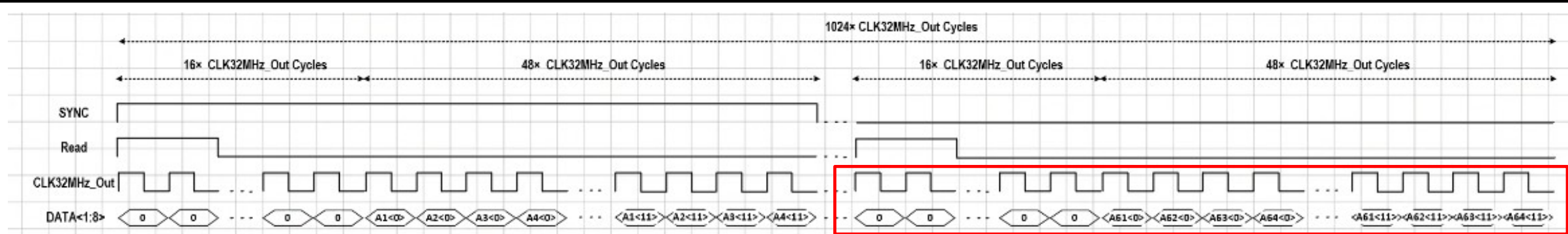
Communication Protocol



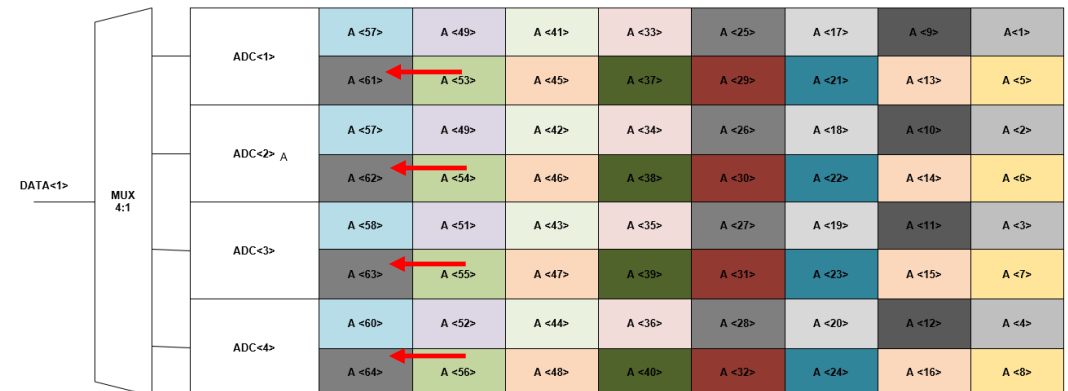
- **1 Frame = 1024 clock cycles**
- **Within each frame, there is one Sync and 16 Read**
- **Sync, signals when the FPGA should start streaming the data → aligns the channel map**
- **Read signals a new amplifier is being streamed**
 - Within each Read period, DATA streams information about 4 amplifiers (from 4 different ADCs which share the same data line)
 - They all have the same location in a group of 16



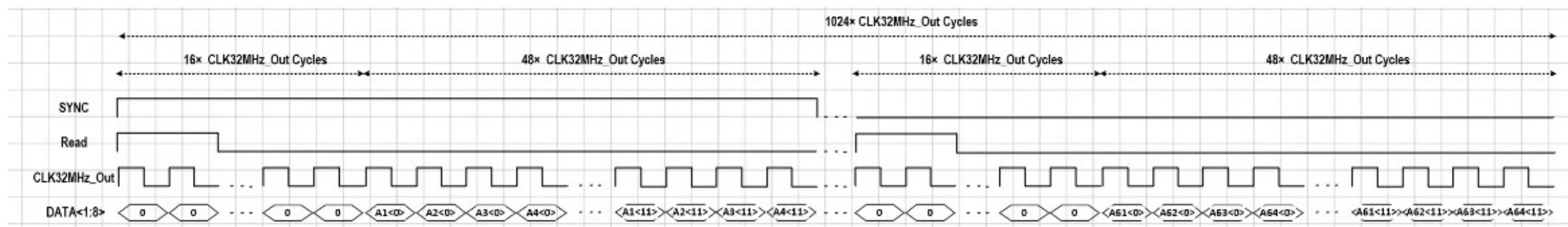
Communication Protocol



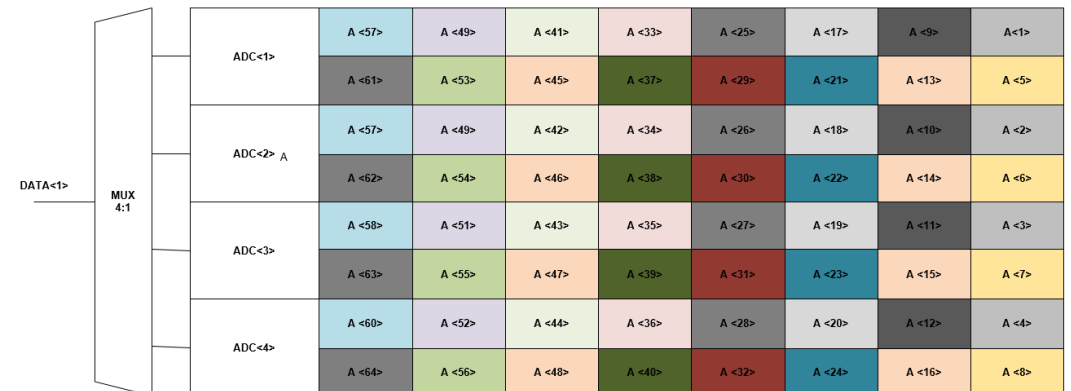
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Communication Protocol

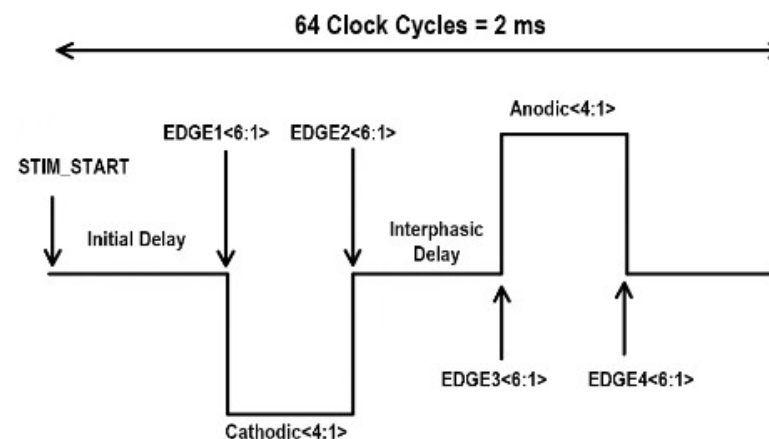
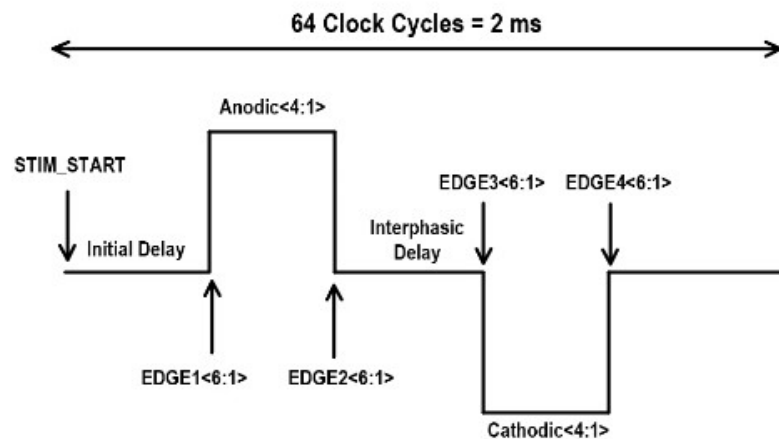
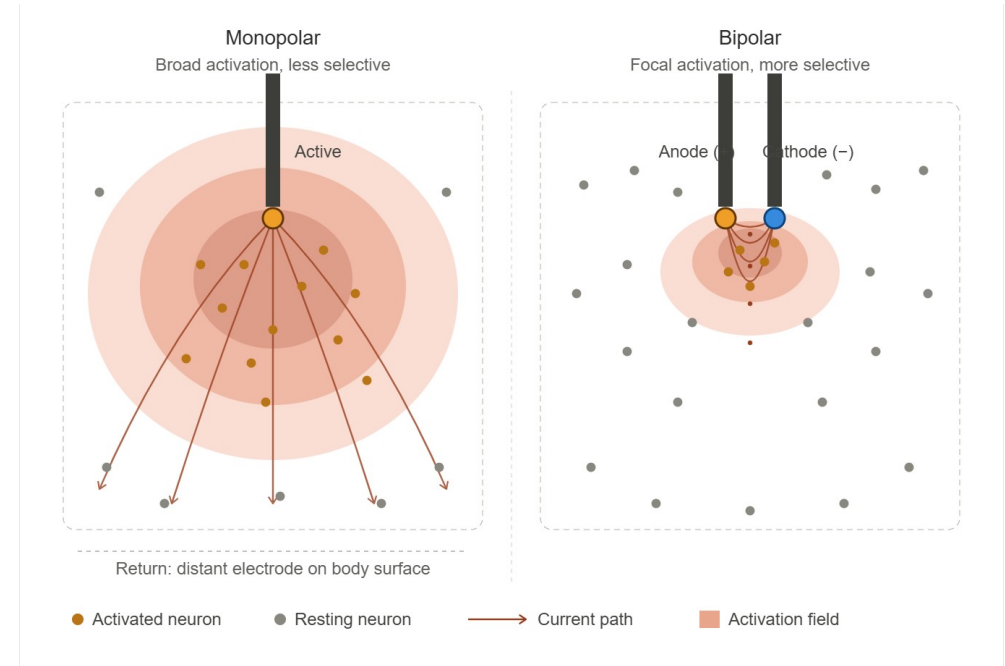


- **1 Frame = 1024 clock cycles**
- **Within each frame, there is one Sync and 16 Read**
- **Sync, signals when the FPGA should start streaming the data → aligns the channel map**
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 - They all have the same location in a group of 16



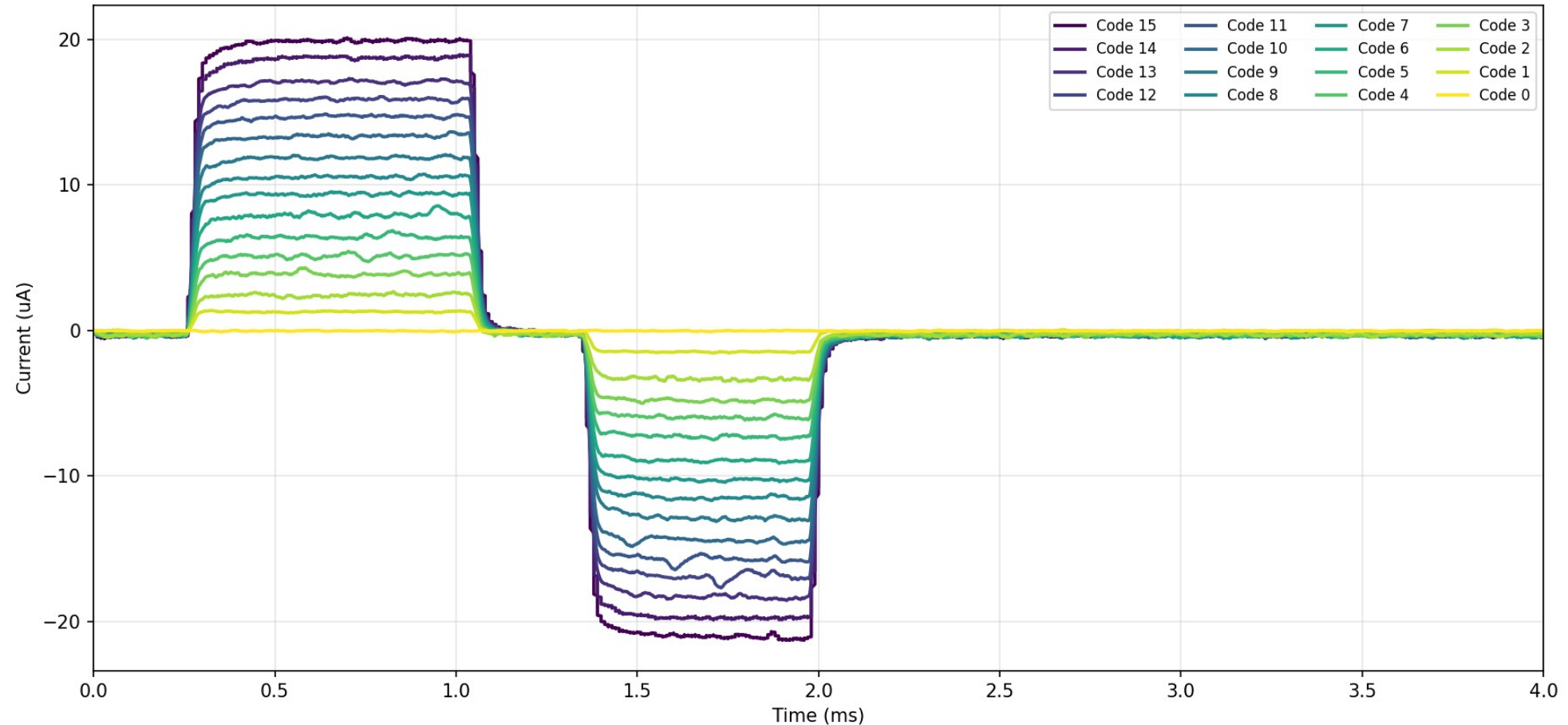
Stimulation Mode

- Biphasic stimulation (monopolar/bipolar)
- Fully programmable arbitrary waveform
- $\pm 10V$ stimulation compliance
- Passive charge balancing



Stimulation Mode - How

- Up to $\sim 22.5\mu\text{A}$ (4-bit control) with a resolution of $\sim 1.5\mu\text{A}$



Stimulation Mode - How

- There are 64 stimulator in each side, left and right

Recording / Chip

Recording: **ON** Start recording Reset amplifier Stop recording

Left Gain: Right Gain: Chip settings... Program gains

Sent 0x30 to setChipTest.exe Program gains sends gains + DACs to the chip via the SPI stream, RUN sends them too.

Stimulation

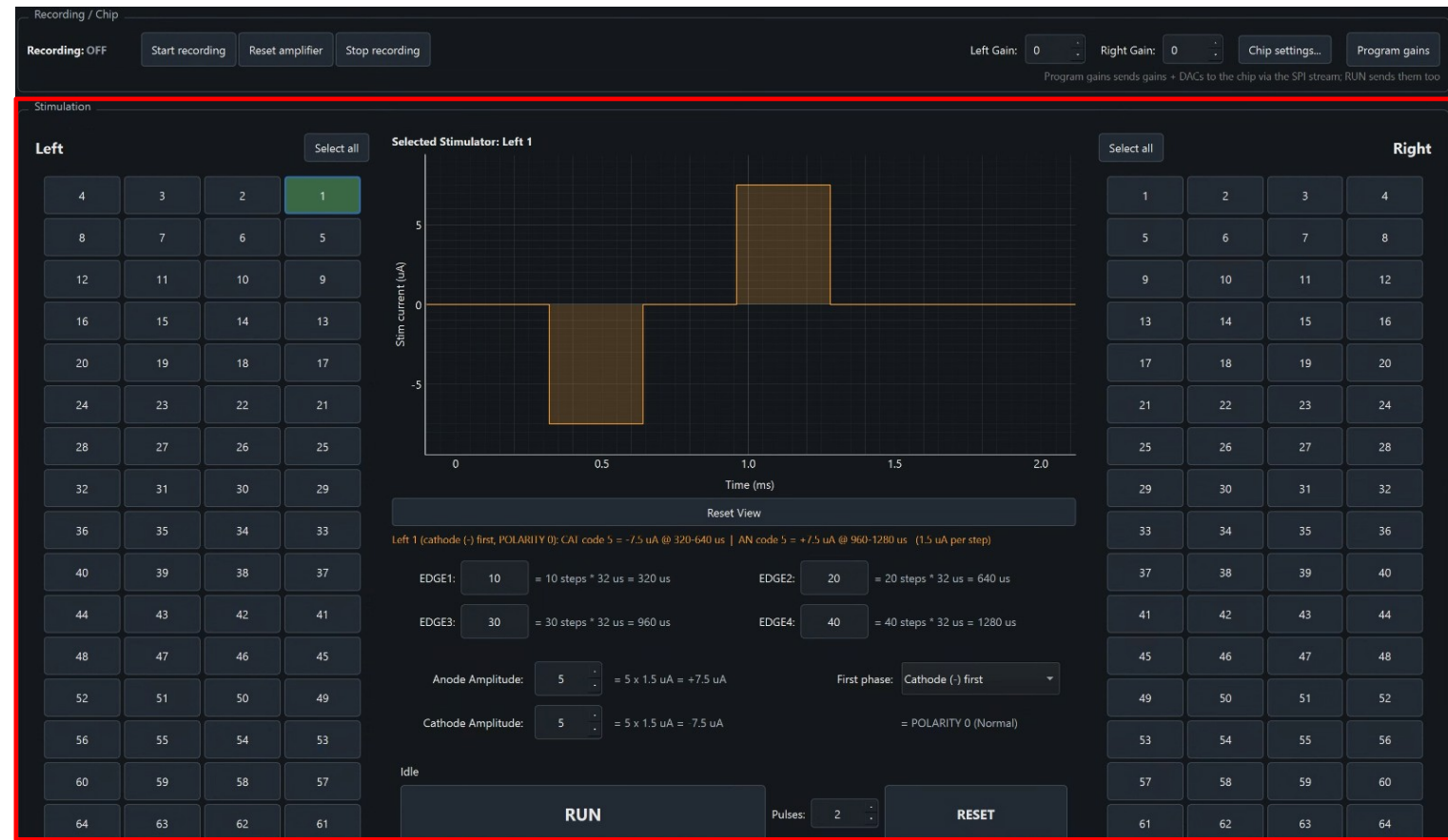
Left Select all Selected Stimulator: none Select all Right

4	3	2	1
8	7	6	5
12	11	10	9
16	15	14	13
20	19	18	17
24	23	22	21
28	27	26	25
32	31	30	29
36	35	34	33
40	39	38	37
44	43	42	41
48	47	46	45
52	51	50	49
56	55	54	53
60	59	58	57
64	63	62	61

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20
21	22	23	24
25	26	27	28
29	30	31	32
33	34	35	36
37	38	39	40
41	42	43	44
45	46	47	48
49	50	51	52
53	54	55	56
57	58	59	60
61	62	63	64

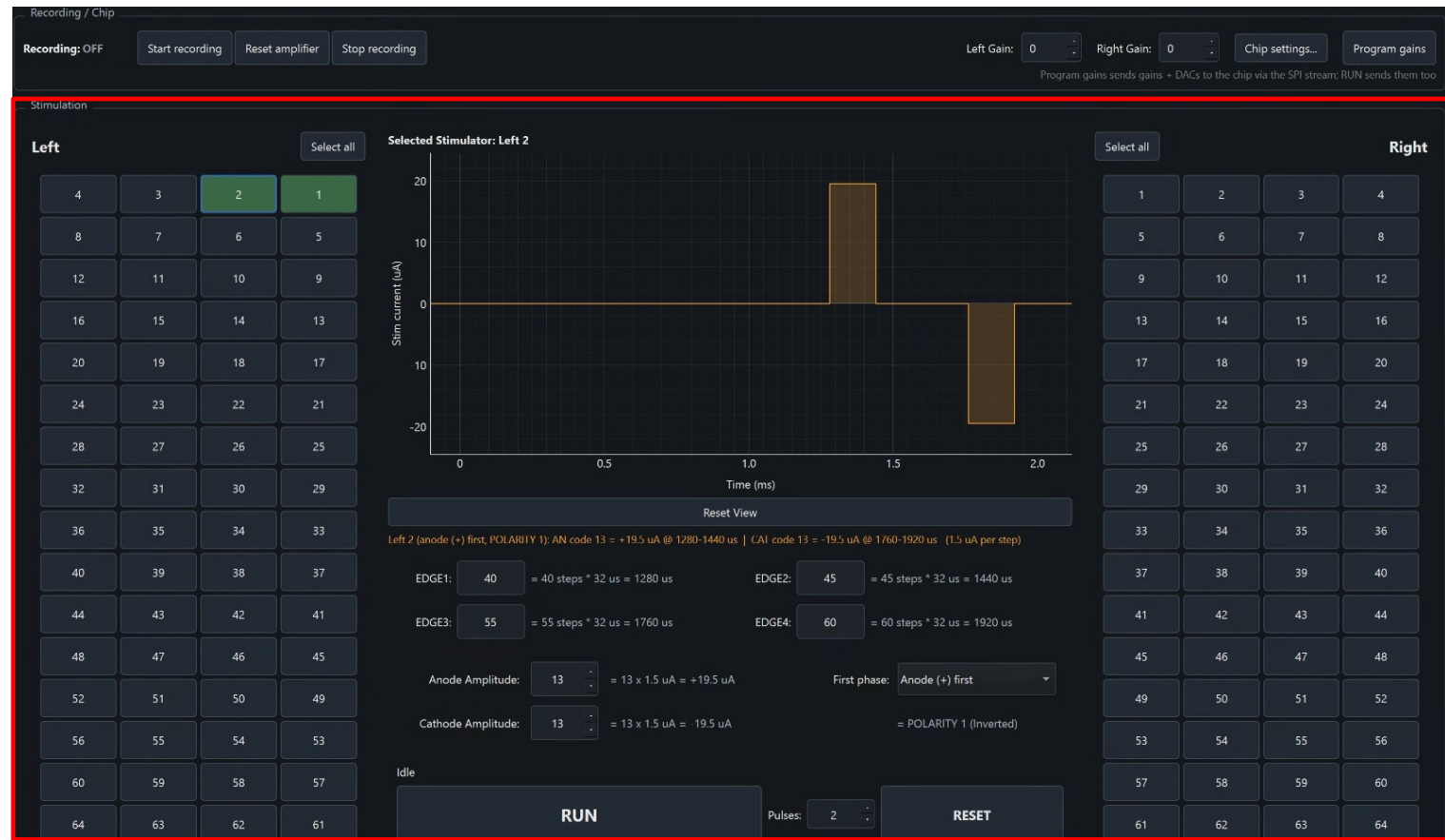
Stimulation Mode - How

- There are 64 stimulator in each side, left and right
- Select the stimulator
 - Fill the edge information based
 - Fill the anode and cathode information
 - Polarity
- Stimulators can have different setting



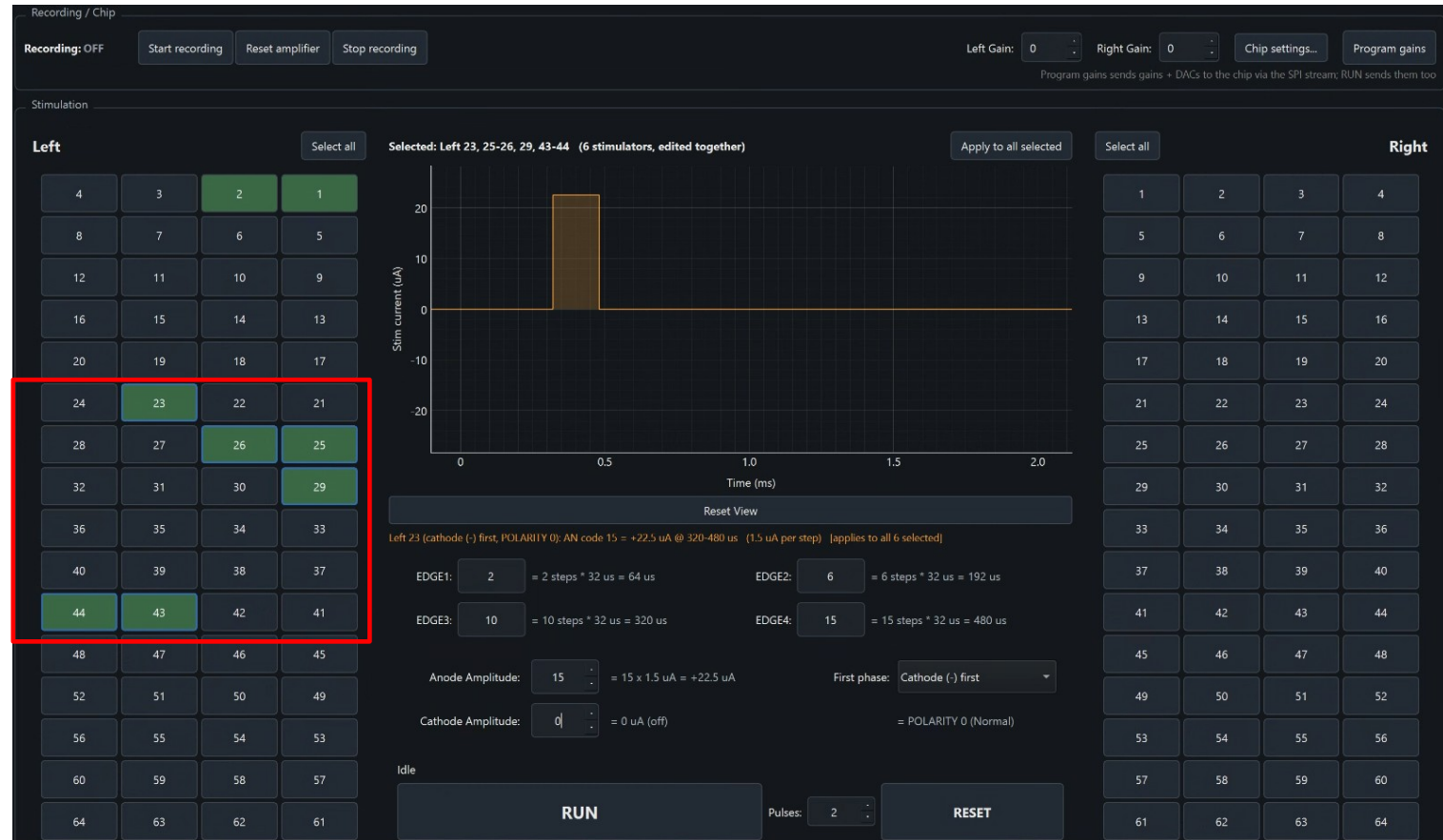
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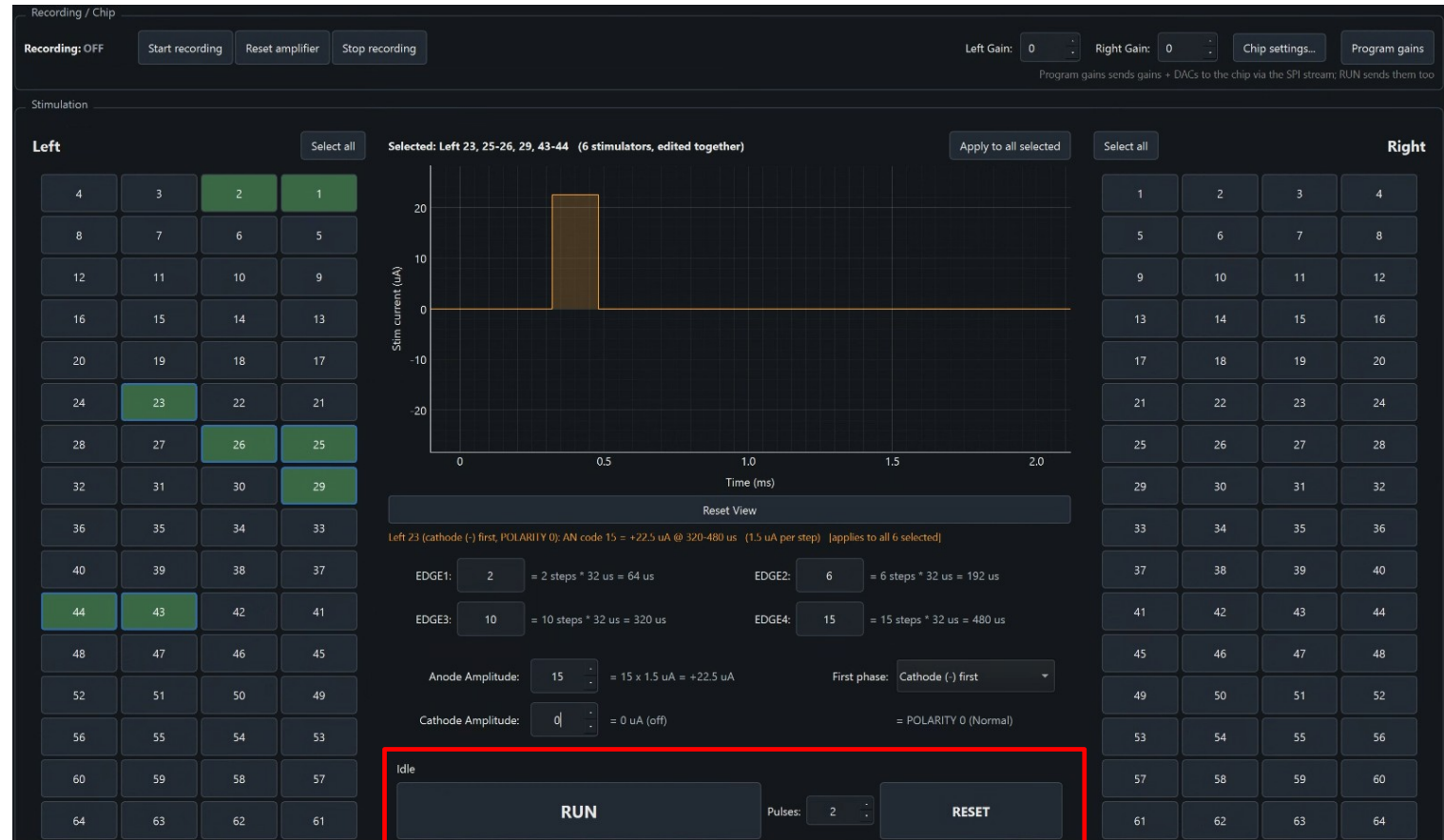
Stimulation Mode - How

- There are 64 stimulator in each side, left and right
- Select the stimulator
 - Fill the edge information based
 - Fill the anode and cathode information
 - Polarity
- Stimulators can have different setting
- Can program multiple of these stimulators at the same time



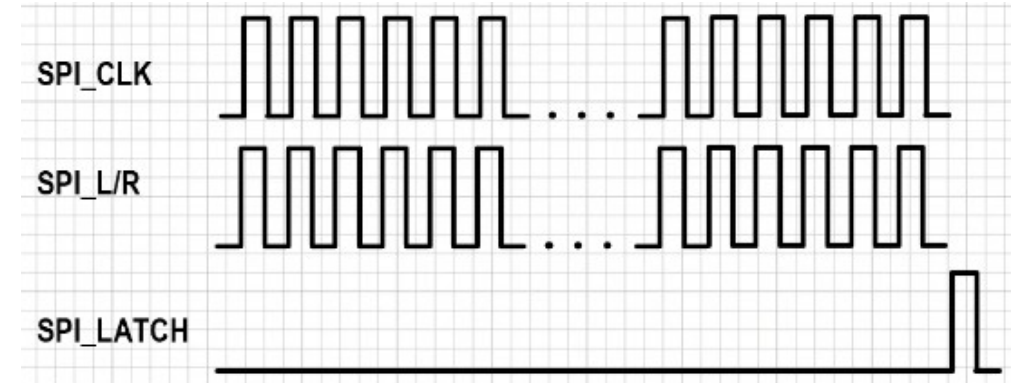
Stimulation Mode - How

- There are 64 stimulator in each side, left and right
- Select the stimulator
 - Fill the edge information based
 - Fill the anode and cathode information
 - Polarity
- Stimulators can have different setting
- Can program multiple of these stimulators at the same time
- Last we select the number of pulses we want to send (1-254) and click RUN



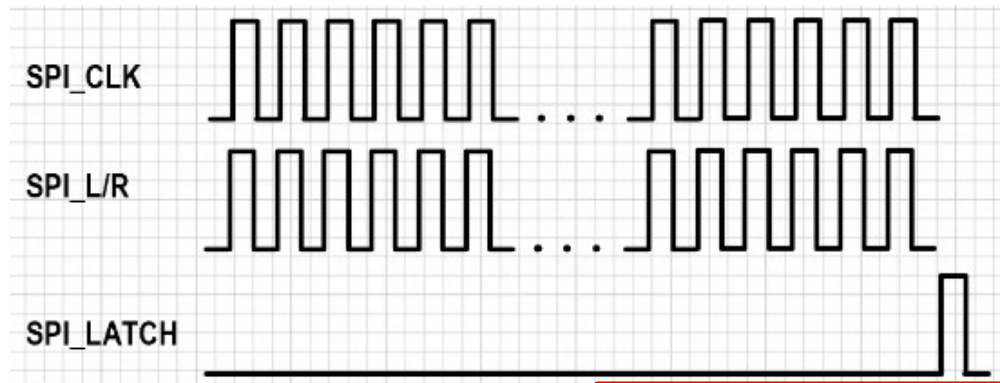
What happens in the background

- **Firstly, an SPI excel file is loaded to the FPGA which then sends the data to the ASIC**
 - The SPI has 4 data pins <CLK, L, R, LATCH>
 - L and R contains information for the stimulators programming and the gain of the amplifiers.



What happens in the background

- Firstly, an SPI excel file is loaded to the FPGA which then sends the data to the ASIC
 - The SPI has 4 data pins <CLK, L, R, LATCH>
 - L and R contains information for the stimulators programming and the gain of the amplifiers.
 - Another patch reads the data in these two excel files and generates the right SPI stream file that downloads to the FPGA



Chip Configuration - 12 bits appended to DR stream
 Each field is a 3-bit value (0..7). Bit<3>=MSB, Bit<1>=LSB.
 Transmission order: DR_DAC<3..1>, AR_DAC<3..1>, Gain_Right<3..1>, Gain_Left<3..1>

FIELD	VALUE	NOTES
Gain_Left	1	3-bit, Gain_Left<3:1>
Gain_Right	1	3-bit, Gain_Right<3:1>
AR_DAC	0	3-bit, AR_DAC<3:1>
DR_DAC	0	3-bit, DR_DAC<3:1> (DR_DAC<3> = SPI<12>, sent first)

duration - 64-Channel System

STIM_ID	ENABLED	EDGE1	EDGE2	EDGE3	EDGE4	CATHODIC_AMP	ANODIC_AMP	POLARITY	NOTES
1	1	0	0	0	0	0	0	1	
2	1	0	0	0	0	0	0	1	
3	1	0	0	0	0	0	0	1	
4	1	0	0	0	0	0	0	1	
5	1	0	0	0	0	0	0	1	
6	1	0	0	0	0	0	0	1	
7	1	0	0	0	0	0	0	1	
8	1	0	0	0	0	0	0	1	
9	1	0	0	0	0	0	0	1	
10	1	0	0	0	0	0	0	1	
11	1	0	0	0	0	0	0	1	
12	1	0	0	0	0	0	0	1	
13	1	0	0	0	0	0	0	1	
14	1	0	0	0	0	0	0	1	
15	1	0	0	0	0	0	0	1	
16	1	0	0	0	0	0	0	1	
17	1	0	0	0	0	0	0	1	
18	1	0	0	0	0	0	0	1	
19	1	0	0	0	0	0	0	1	
20	1	0	0	0	0	0	0	1	
21	1	0	0	0	0	0	0	1	
22	1	0	0	0	0	0	0	1	
23	1	0	0	0	0	0	0	1	
24	1	0	0	0	0	0	0	1	
25	1	0	0	0	0	0	0	1	
26	1	0	0	0	0	0	0	1	
27	1	0	0	0	0	0	0	1	
28	1	0	0	0	0	0	0	1	
29	1	0	0	0	0	0	0	1	
30	1	0	0	0	0	0	0	1	
31	1	0	0	0	0	0	0	1	
32	1	0	0	0	0	0	0	1	
33	1	0	0	0	0	0	0	1	
34	1	0	0	0	0	0	0	1	
35	1	0	0	0	0	0	0	1	
36	1	0	0	0	0	0	0	1	
37	1	0	0	0	0	0	0	1	
38	1	0	0	0	0	0	0	1	
39	1	0	0	0	0	0	0	1	
40	1	0	0	0	0	0	0	1	
41	1	0	0	0	0	0	0	1	
42	1	0	0	0	0	0	0	1	
43	1	0	0	0	0	0	0	1	
44	1	0	0	0	0	0	0	1	
45	1	0	0	0	0	0	0	1	
46	1	0	0	0	0	0	0	1	

bit_index	DL	DR
1	0	0
2	0	0
3	0	0
4	0	0
5	0	0
6	0	0
7	0	0
8	0	0
9	0	0
10	0	0
11	0	0
12	0	0
13	0	0
14	0	0
15	0	0
16	0	0
17	0	0
18	0	0
19	0	0
20	0	0
21	0	0
22	0	0
23	0	0
24	0	0
25	0	0
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29	0	0
30	0	0
31	0	0
32	0	0
33	0	0

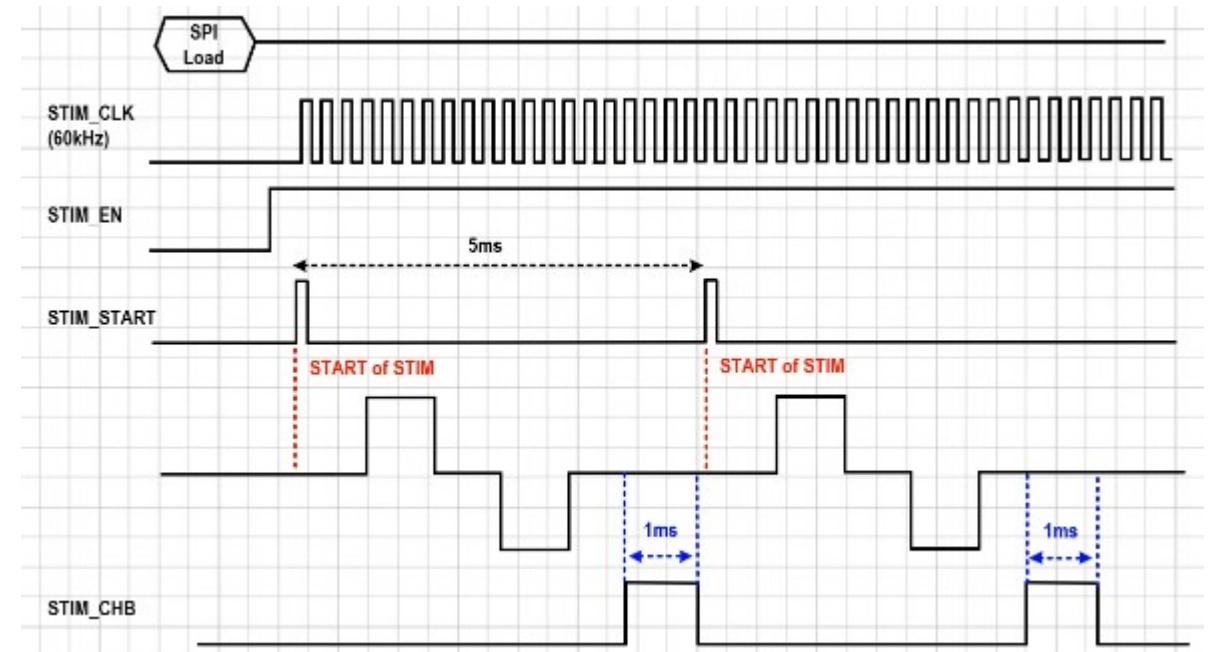
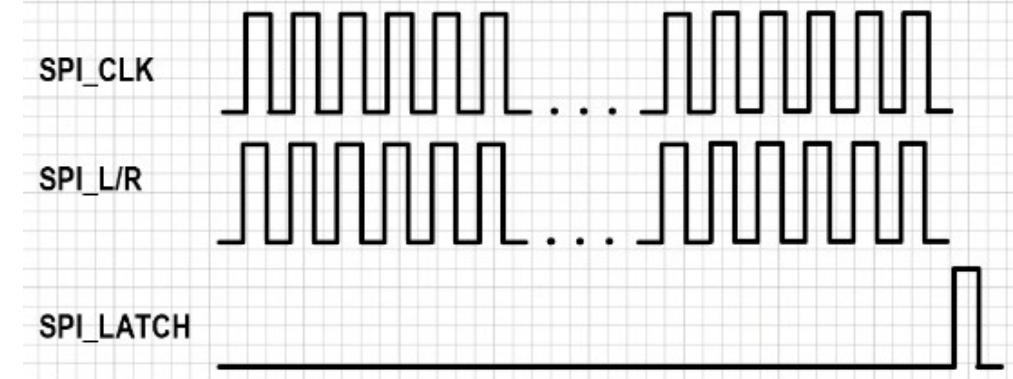
Gain Config

STIM Config

SPI Stream

What happens in the background

- **Firstly, an SPI excel file is loaded to the FPGA which then sends the data to the ASIC**
 - The SPI has 4 data pins <CLK, L, R, LATCH>
 - L and R contains information for the stimulators programming and the gain of the amplifiers.
 - Another patch reads the data in these two excel files and generates the right SPI stream file that downloads to the FPGA
- **How does the stimulator start working?**
 - Firstly, we enable the stims via STIM_EN
 - Then we send a short pulse to trigger the stims → STIM_START
 - After 64 STIM_CLK cycles, we enable STIM_CHB which does the charge balancing



Different Stim patterns each cycle?

- Firstly, an SPI excel file is loaded to the FPGA which then sends the data to the ASIC
 - The SPI has 4 data pins <CLK, L, R, LATCH>
 - L and R contains information for the stimulators programming and the gain of the amplifiers.
 - Another patch reads the data in these two excel files and generates the right SPI stream file that downloads to the FPGA
- How does the stimulator start working?
 - Firstly, we enable the stims via STIM_EN
 - Then we send a short pulse to trigger the stims → STIM_START
 - After 64 STIM_CLK cycles, we enable STIM_CHB which does the charge balancing
- Different stimulation sequence is possible as long as we can generate the timing diagram shown in the right

